

Ph.D. Scholar in Computer Science
Marie Curie Fellow

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Collaborative Networking Lab
Department of Computer Science
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Education

- 2015 – **Ph.D. Scholar** **University of Helsinki**
Finland
– Supervised by *Prof. Dr. Jussi Kangasharju*
– Collaborative Networking (CoNe) research group
- 2013 – 2015 **M.Tech. Computer Science & Engineering** **IIIT Delhi**
India
– Specialization in *Mobile Computing*
– CGPA: 8.5/10
– Nominated for *Best Thesis Award*
- 2009 – 2013 **B.Tech. Computer Engineering** **Punjabi University**
India
– First Division *with Distinction*
– Ranked amongst *Top-10 batch performers*

Skills

- Research Interests **Edge clouds, Cloud computing, Internet-of-Things, Multipath TCP, Information-Center Networks, Distributed Systems**
- Programming **Python, Java, HTML5, C, C++, BASH, L^AT_EX**
- Tools & Technologies **Linux kernel programming, Network Simulator (NS-3), MATLAB, SPARK, Docker, GRAPHLAB, TENSORFLOW**
- Languages **English Expert Hindi Expert Punjabi Expert Finnish Beginner**

Research Experience

- Ph.D. **University of Helsinki** **Helsinki, Finland**
Sept 2015 – June 2019 (expected)
Supervisor: *Prof. Dr. Jussi Kangasharju*
– Research in operation, deployment & networking in edge clouds
– Jointly funded by EU FP7 project – CleanSky ITN & Finnish Academy of Sciences project – AIDA
- Visiting Researcher **University of Göttingen** **Göttingen, Germany**
Jan – May 2018
Host: *Prof. Dr. Xiaoming Fu*
– Research in optimal placement of edge servers and virtualized service functions
– Funded by EU FP7 project – CleanSky ITN

Research Experience (continued)

	■ NEC research labs, Europe Host: <i>Dr. Fabian Schneider</i> – Research in deploying network functions in edge clouds – Funded by EU FP7 project – CleanSky ITN	Heidelberg, Germany Nov 2016 – Mar 2017
Project Scientist	■ Indian Inst. of Technology (IIT) Supervisor: <i>Dr. Vinay Ribeiro</i> – Research & Development in "Structured sharing of resources in a Device-to-Device (D2D) Network" – Funded by Intel Labs, USA	New Delhi, India June – Aug 2015
M.Tech. (with Thesis)	■ Indraprastha Inst. of Information Tech. (IIIT) Supervisor: <i>Dr. Pushpendra Singh</i> – Research in checkpointing user-applications in Information-centric networks – Awarded GATE scholarship by Govt. of India	New Delhi, India 2013 – 2015
Research Intern	■ Central Scientific Instruments Org. (CSIO) Supervisor: <i>Dr. Sudesh K. Mittal</i> – Research in landslide prediction model using image processing and interpolation – Collaboration with Indian Space Research Org. (ISRO)	Chandigarh, India Jan – July 2013

Teaching Experience

Guest Instructor	■ Cloud & Edge Computing (University of Helsinki)	2019
Teaching Assistant	■ Distributed Data & Infrastructures (University of Helsinki) ■ Distributed Systems (University of Helsinki) ■ Cellular Data Networks (IIIT Delhi)	2017, 2018 2016 2015

Positions of Responsibility

Shadow PC	■ ACM Internet Measurement Conference (IMC)	2019
External Reviewer	■ IEEE ICC NGNI Symposium ■ ACM Information Centeric Networking Conference (ICN)	2019 2017
Jury Organizer	■ HackEdge hackathon organized by CEACS ■ 3rd International CleanSky Conference ■ 2nd International CleanSky Summer School	2017 2017 2016
Web-Admin	■ 2nd International CleanSky Summer School	2016
Head Representative	■ Design Team of IIITD Research Showcase ■ M.Tech. 2013-15 batch at IIIT Delhi ■ Post-Graduate Disciplinary Action Committee at IIIT Delhi	2015 2013, 2014 2013, 2014

Awards & Achievements

- 1 Awarded [Microsoft Research travel grant](#) for ACM MOBICOM 2018 held in New Delhi, India
- 2 Awarded EU FP7 [Marie Curie Initial Training Network \(ITN\)](#) funding to support Ph.D. studies
- 3 Invited to attend [South Asia Workshop On Research Frontiers In Computing](#) with full-fellowship held at National University of Singapore (NUS), Singapore in May, 2014
- 4 [All-India percentile 98.57](#) in Graduate Aptitude Test Examination (GATE) – 2014 amongst ~2 million candidates
- 5 Awarded [GATE scholarship](#) by Govt. of India from 2013 – 2015
- 6 [Recognition Award](#) by Director, IIIT Delhi for exceptional service to Institute from 2013 – 2015
- 7 Graduated Bachelors with [1st Division with Distinction](#) and ranked amongst top 10 students
- 8 Awarded [Certificate of Merit](#) by Govt. of India for being [All-India Topper](#) in Mathematics in All-India Secondary School Examination in 2007

Research Projects

AIDA	■ Adaptive and Intelligent Data – Objective : Introducing learning and intelligence in networks – PI : <i>Prof. Dr. Jussi Kangasharju</i> (Univ. of Helsinki) <i>Prof. Dr. Aristides Gionis</i> (Aalto Univ.)	Academy of Finland 2018–2021
BCDC	■ Bright Clouds Dark Clouds www.bcdcenergia.fi – Objective : Digitalizing utilization of renewable energy via Cloud – PI : <i>Prof. Dr. Rauli Svento</i> (Univ. of Oulu)	Academy of Finland 2016–2020
CleanSky	■ CleanSky Initial Training Network www.cleansky-itn.eu – Marie Curie Framework Program (FP) 7 Project – Objective : Innovation in several areas of cloud and edge computing – Collaboration : Nine university and industry partners in Europe, USA & China – PI : <i>Prof. Dr. Xiaoming Fu</i> (Univ. of Göttingen)	European Union 2014–2018

Research Contribution

Journal Publications

2018	■ Placing it right!: Optimizing Energy, Processing, & Transport in Edge-Fog clouds N. Mohan & J. Kangasharju International Journal on <i>Annals of Telecommunication</i> , Springer, 2018 Paper: https://doi.org/10.1007/s12243-018-0649-0	Springer
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Conference & Workshop Publications

2019	■ [Manuscript] Dropping off the Edge!: Pruning Edge Research with Latency Shears N. Mohan , A. Zavodovski , W. Wong , P. Zhou & J. Kangasharju 18 th ACM Workshop on <i>Hot Topics in Networks</i>	HotNets '19
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Research Contribution (continued)

- [In Review] Is two greater than one?: Analyzing Multipath TCP over dual-LTE in the Wild IMC '19
N. Mohan, T. Shreedhar, A. Zavodovski, J. Kangasharju & S.K. Kaul
ACM Internet Measurement Conference

- Open Infrastructure for Edge: A Distributed Ledger Outlook HotEdge '19
A. Zavodovski, N. Mohan, W. Wong & J. Kangasharju
USENIX ATC workshop on *Hot Topics in Edge Computing*

- DeCloud: Truthful Decentralized Double Auction for Edge Clouds ICDCS '19
A. Zavodovski, S. Bayhan, N. Mohan, P. Zhou, W. Wong & J. Kangasharju
39th International Conference on Distributed Computing Systems

- ExEC: Elastic Extensible Edge Cloud EdgeSys '19
A. Zavodovski, N. Mohan, S. Bayhan, W. Wong & J. Kangasharju
ACM EuroSys workshop on *Edge Systems, Analytics & Networking*
Paper: <https://doi.org/10.1145/3301418.3313941>
Slides: <http://bit.ly/2YGZ3jD>

- 2018 ■ ICON: Intelligent Container Overlays HotNets '18
A. Zavodovski, N. Mohan, S. Bayhan, W. Wong & J. Kangasharju
17th ACM Workshop on Hot Topics in Networks
Paper: <https://doi.org/10.1145/3286062.3286065>
Slides: <https://bit.ly/2UMcAUE> Video: <https://youtu.be/BNmF0gTV1R4>

- Anveshak: Placing Edge Servers In The Wild MECOMM '18
N. Mohan, A. Zavodovski, P. Zhou & J. Kangasharju
ACM SIGCOMM workshop on *Mobile Edge Communications*
Paper: <https://doi.org/10.1145/3229556.3229560>
Slides: <https://bit.ly/2W9JWx3> Video: <https://youtu.be/8wYjQr80iwI>

- QAware: A Cross-Layer Approach to MPTCP Scheduling IFIP Networking '18
T. Shreedhar, N. Mohan, S.K. Kaul & J. Kangasharju
IFIP Networking Conference, Zürich. May 2018
Paper: <https://bit.ly/2CtTITj>
Open Source: <https://bit.ly/2Jod1Du>

- MUTE: Multi-Tier Edge networks CrossCloud '18
A. Silvestro*, N. Mohan*, J. Kangasharju, F. Schneider & X. Fu [**joint authors*]
ACM EuroSys workshop on *CrossCloud Infrastructures & Platforms*
Paper: <https://doi.org/10.1145/3195870.3195871>
Slides: <https://bit.ly/2CxCjct>

- 2017 ■ Managing Data in Computational Edge Clouds MECOMM '17
N. Mohan, P. Zhou, K. Govindaraj & J. Kangasharju
ACM SIGCOMM workshop on *Mobile Edge Communications*
Paper: <https://doi.org/10.1145/3098208.3098212>
Slides: <https://bit.ly/2HJB5yn> Video: <https://youtu.be/akJtWokhrj4>

Research Contribution (continued)

- 2016  Edge-Fog Cloud: A Distributed Cloud for Internet of Things Computations CIoT '16
N. Mohan & J. Kangasharju
IEEE conference on *Cloudification of the Internet of Things*
Paper: <https://doi.org/10.1109/CIOT.2016.7872914>
Slides: <https://bit.ly/2Hwdrpz>

Posters & Demos

- 2018  Redesigning MPTCP for Edge Clouds MobiCom '18
N. Mohan, T. Shreedhar, A. Zavodovski, O. Waltari, J. Kangasharju & S.K. Kaul
ACM conference on *Mobile Computing and Networking*
Paper: <https://doi.org/10.1145/3241539.3267738>
Poster: <https://bit.ly/2TNfUCB> SRC slides: <https://bit.ly/2Fg1iSk>
- 2017  Grouping Computational Data in Resource Caches of Edge-Fog Cloud CrossCloud '17
N. Mohan, P. Zhou, K. Govindaraj, & J. Kangasharju
ACM EuroSys workshop on *CrossCloud Infrastructures & Platforms*
Paper: <https://doi.org/10.1145/3069383.3069391>
Poster: <https://bit.ly/20h2qsJ>
- 2014  CCNCheck: Enabling Efficient Checkpointed Applications in Content Centric Networks CCNxCon '14
N. Mohan & P. Singh
PARC International *Content Centric Networking conference*
Paper: <https://bit.ly/2CtWqIy>

Thesis

- 2015  Enabling Sender-initiated Distributed Applications and Checkpointing in Content Centric Networks M.Tech.
Best Thesis Nominee
Nitinder Mohan
Submitted at: *IIIT Delhi, India. May 2015*
Examiner: *Dr. Vinay Ribeiro (IIT Delhi) & Dr. Vikram Goyal (IIIT Delhi)*
Paper: <https://bit.ly/2TWyKAd>

Talks

- 2018  Providing Networking at Edge CleanSky Workshop '18
4th International CleanSky Summer School
Host: Prof. Dr. Jussi Kangasharju
University of Helsinki, Finland
- 2017  Scaling computational clouds to network edge CleanSky Summer School '17
3rd International CleanSky Summer School
Host: Prof. Dr. Dan Li
Tsinghua University, Beijing, China
-  Filling Edge-Fog cloud cover: All power to robots! CleanSky Workshop '17
3rd International CleanSky Workshop
Host: Prof. Dr. Xiaoming Fu
University of Göttingen, Germany

Research Contribution (continued)

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| 2016 | <ul style="list-style-type: none"> ■ <i>MiddleBox Broker</i>: Enabling in-network functions in secured traffic
2nd International CleanSky Workshop
Host: Prof. Dr. Yuming Jiang
NTNU Trondheim, Norway ■ <i>Edge-Fog cloud</i>: Scaling IoT computations on the Edge
Aware Machine-to-Machine Communication
GI-Dagstuhl Seminar 16353
Dagstuhl, Germany ■ Computing IoT data on <i>Edge-Fog cloud</i>
2nd International CleanSky Summer School
Host: Prof. Dr. Jussi Kangasharju
University of Helsinki, Finland ■ Simplifying IoT computation in the “Cloud”
1st International CleanSky Workshop
Host: Dr. Fabian Schneider
NEC Europe Labs, Heidelberg, Germany | <div>CleanSky Workshop '16</div> <div>GI-Dagstuhl '16</div> <div>CleanSky Summer School '16</div> <div>CleanSky Workshop '16</div> |
| 2015 | <ul style="list-style-type: none"> ■ Scaling eScience computational-intensive cloud
1st International CleanSky ITN Summer School
Host: Prof. Dr. Xiaoming Fu
University of Göttingen, Germany | <div>CleanSky Summer School '15</div> |

Conference Participation

International Conferences

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| 2018 | <ul style="list-style-type: none"> ■ ACM MOBICOM ■ ACM SIGCOMM ■ IFIP Networking ■ EuroSys | <div>New Delhi, India</div> <div>Budapest, Hungary</div> <div>Zürich, Switzerland</div> <div>Porto, Portugal</div> |
| 2017 | <ul style="list-style-type: none"> ■ Networked Systems ■ EuroSys | <div>Göttingen, Germany</div> <div>Belgrade, Serbia</div> |
| 2016 | <ul style="list-style-type: none"> ■ ACM SIGCOMM ■ IEEE Cloudification of IoT | <div>Florianopolis, Brazil</div> <div>Paris, France</div> |

Seminars/Summer Schools

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| 2017 | <ul style="list-style-type: none"> ■ 3rd CleanSky ITN Summer School
nasp.cs.tsinghua.edu.cn/CleanSky/ | <div>Beijing, China</div> |
| 2016 | <ul style="list-style-type: none"> ■ [GI-Dagstuhl] Aware-to-aware machine communication
www.dagstuhl.de/en/program/calendar/evhp/?seminr=16353 ■ 2nd CleanSky ITN Summer School
cleanskysummer.cs.helsinki.fi/ ■ Mobile Services and Edge Computing Workshop
www.cs.helsinki.fi/group/close/edge-computing-2016/ | <div>Dagstuhl, Germany</div> <div>Helsinki, Finland</div> <div>Organizer Web-Chair</div> <div>Helsinki, Finland</div> |
| 2015 | <ul style="list-style-type: none"> ■ 1st CleanSky ITN Summer School
cleanskysummer.uni-goettingen.de/ | <div>Helsinki, Finland</div> |
| 2014 | <ul style="list-style-type: none"> ■ South Asia Workshop On Research Frontiers In Computing
www.comp.nus.edu.sg/programmes/pg/workshops/southasia/ | <div>Singapore</div> |

Open Source Contributions

- [rpi-mptcp](#), A tool for measuring MPTCP performance over LTE links on Raspberry Pi
- [anveshak](#), A tool which calculates optimal deployment of cellular Edge servers in an area
- [ns3-qaware](#), ns3 implementation of MPTCP QAware scheduler
- [qaware](#), Linux kernel implementation of MPTCP QAware scheduler
- [efsim](#) A Python-based simulator for Edge cloud networks
- [ccnac](#), A tool to enable application checkpointing in Information-centric networks

Memberships

- Association of Computing Machinery (ACM) Student Member
- Institute of Electrical and Electronics Engineers (IEEE) Student Member
- SIGCOMM Membership
- EUROSYS Membership
- Indian Society of Technical Education (ISTE) Student Member

References

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